# Homework 3 Code

Matthew Schoen

2026-05-28

## Part 1: Set Up

```
# Reading in necessary packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)

# Storing kelp data as specified object
kelp <- read_csv(
  here("data", "temp-kelp.csv"))

# Storing personal data as specificed object
step_count <- read_csv(
  here("data", "MyData_HW3_Format.csv"))
```

## Part 2: Problems

### Problem 1 - Giant Kelp Fronds

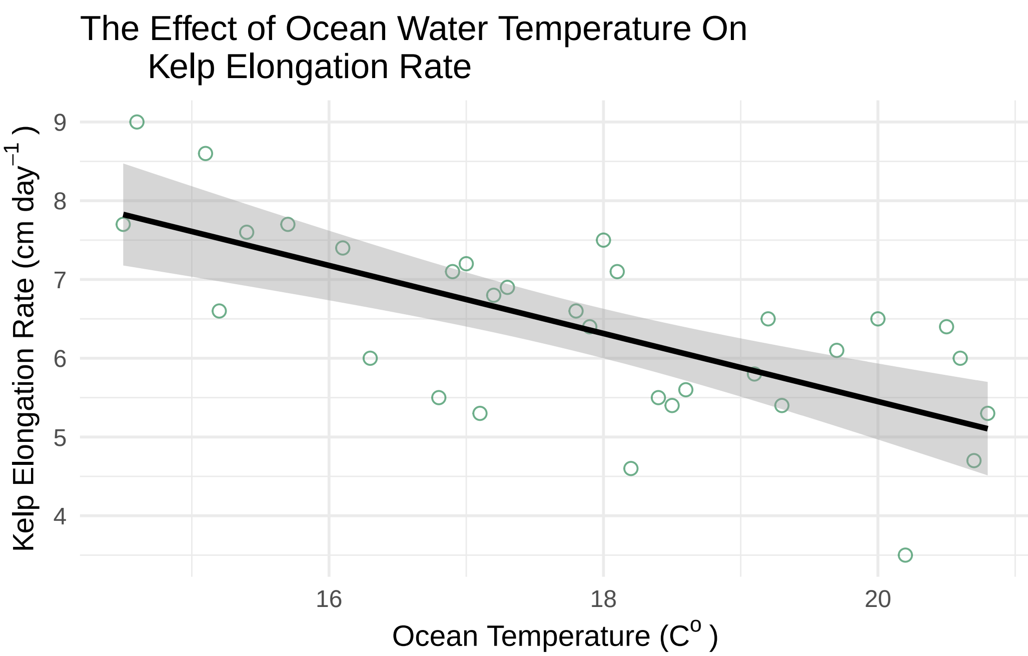
#### a. An Appropriate Test

To evaluate the strength of the relationship between temperature ($C^o$) and giant kelp frond elongation rate ($cmday^-1$), a Pearson correlation test and a Spearman rank correlation test would both be suited to use. The Pearson correlation test will evaluate whether there is a linear relationship between the two continuous variables while also assuming a normal distribution and linear relationship and the Spearman-rank correlation test will evaluate if a relationship exists between the two variables without requiring a normal distribution. As both ocean temperature and elongation rate are continuous and quantitative variables, both tests could be used depending on whether the data is normally distributed and linear.

### b. Create a Visualization

```
# Creating a ggplot base layer using kelp dataset
  ggplot(data = kelp,
         mapping = aes(x = temp_c,
                       y = kelp_elong)) +

  # First layer is a scatter plot to visualize individual observations
  geom_point(size = 2,
             alpha = 0.7,
             color = "#2E8B57",
             shape = 21) +
  # Second layer is a line of best fit using a linear regression
  geom_smooth(method = "lm",
              color = "black",
              linewidth = 1
             ) +
  # Using custom theme
  theme_minimal() +
  # Relabeling x & y axes
  labs(x = expression("Ocean Temperature (C"^o~")"),
       y = expression("Kelp Elongation Rate (cm day"^-1~")"),
       title = "The Effect of Ocean Water Temperature On
       Kelp Elongation Rate")
```

```
`geom_smooth()` using formula = 'y ~ x'
```

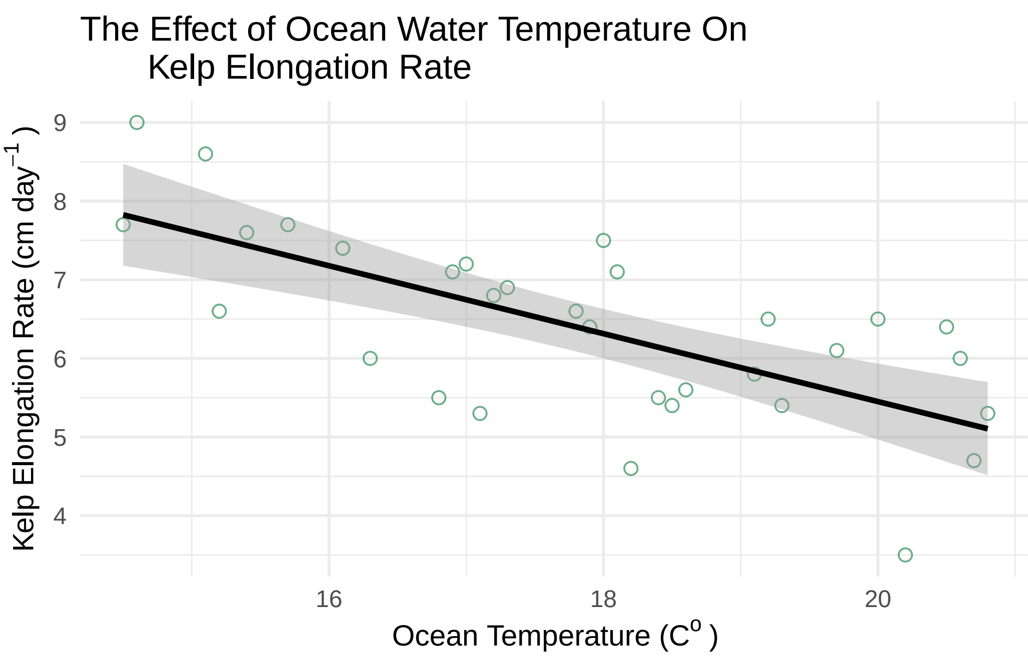



#### c. Check Your Assumptions and Run Your Test

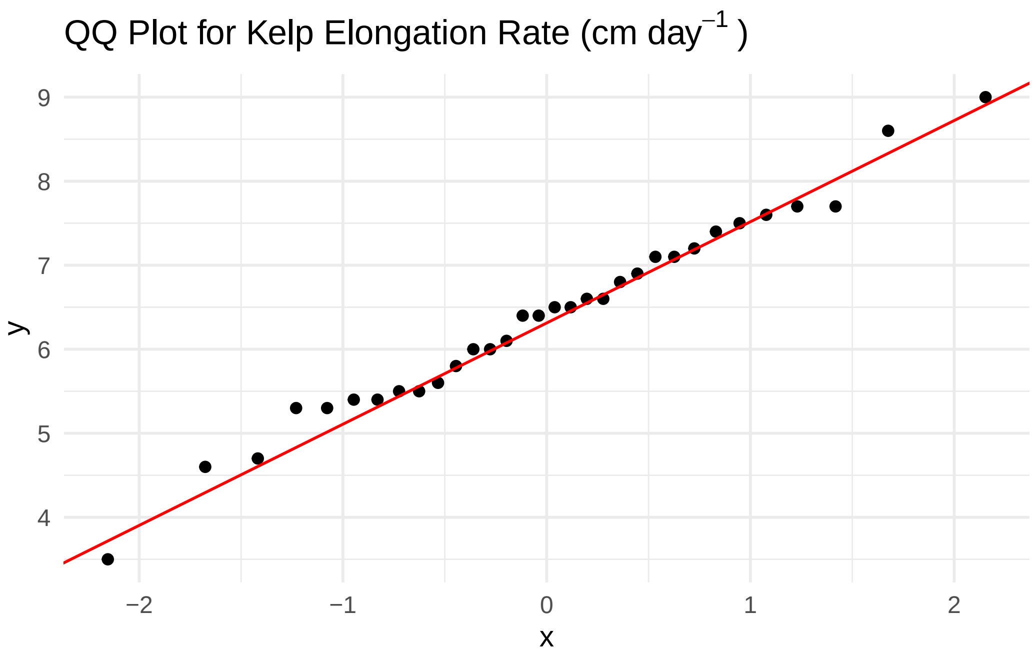
```
# Creating a ggplot base layer using kelp dataset
  ggplot(data = kelp,
         mapping = aes(x = temp_c,
                       y = kelp_elong)) +

  # First layer is a scatter plot to visualize individual observations
  geom_point(size = 2,
             alpha = 0.7,
             color = "#2E8B57",
             shape = 21) +
  # Second layer is a line of best fit using a linear regression
  geom_smooth(method = "lm",
              color = "black",
              linewidth = 1
             ) +
  # Using custom theme
  theme_minimal() +
  # Relabeling x & y axes
  labs(x = expression("Ocean Temperature (C"^o~")"),
```

```
       y = expression("Kelp Elongation Rate (cm day"^-1~")"),
       title = "The Effect of Ocean Water Temperature On
       Kelp Elongation Rate")
```

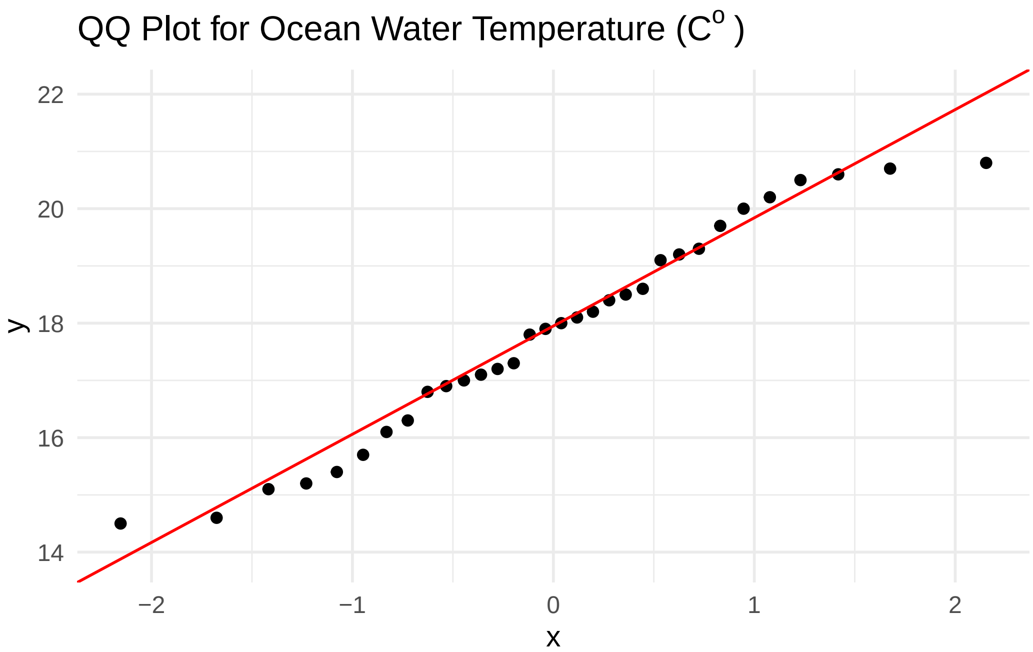
```
`geom_smooth()` using formula = 'y ~ x'
```




```
# ggplot as base layer
  ggplot(data = kelp,
         mapping = aes(sample = kelp_elong)) +
  # first layer is QQ plot
  geom_qq() +
  # second layer is QQ line
  geom_qq_line(color = "red") +
  # updating theme
  theme_minimal() +
  # updating title
  labs(title = expression("QQ Plot for Kelp Elongation Rate (cm day"^-1~")"))
```




```
# ggplot as the base layer
  ggplot(data = kelp,
   mapping = aes(sample = temp_c)) +
  # first layer is the qq plot
  geom_qq() +
  # second layer is the qq plot line
  geom_qq_line(color = "red") +
  # updating theme
  theme_minimal() +
  # updating title
  labs(title = expression("QQ Plot for Ocean Water Temperature (C"^o~")"))
```




**Write-Up:**

To evaluate whether a Pearson correlation test or a Spearman-rank correlation test should be used, I assessed the linearity of the two data in addition to each variable's distribution. Using a scatterplot with a fitted linear regression, it was determined that the data *appears* to have somewhat of a negatively linear relationship where as ocean temperature increases, kelp elongation rate decreases. Using qq plots, it was determined that both Ocean Temperature and Kelp Elongation Rate are normally distributed as data points fall along a liner line of best fit.

```r
# Running a Pearson corrleation test
cor.test(x = kelp$temp_c,
         y = kelp$kelp_elong,
         method = "pearson")
```

```
    Pearson's product-moment correlation

data:  kelp$temp_c and kelp$kelp_elong
t = -5.189, df = 30, p-value = 1.366e-05
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.8359665 -0.4460157
```

```
sample estimates:
       cor
-0.6877489
```

### d. Results Communication

### e. Test Implications

### f. Double Check Your Own Work

## Problem 2 - Personal Data

### a. Updating Your Visualizations

### b. Captions

## Problem 3 - Affective Visualization

### a. Describe An Affective Visualization

### b. Create A Sketch

### c. Make A Draft

### d. Write An Artist Statement

### e. Prep Your Materials

## Problem 4 - Statistical Critique

### a. Revisit and Summarize

### b. Visual Clarity

### c. Aesthetic Clarity

### d. Reccomendations